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A fast data-driven battery capacity estimation method under non-constant current charging and variable temperature

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A R T I C L E I N F O

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A B S T R A C T

Data-driven methods have been widely used in capacity estimation of lithium-ion batteries. Non-constant current charging and variable-temperature operating scenarios are inevitable in real applications. However, existing classical constant current charging based capacity estimation methods may not be applied to such scenarios. To this end, a data-driven method adapted to non-constant current charging is proposed in this work. A single health feature collected within 3 min is combined with Gaussian process regression (GPR) to achieve fast and transferable battery capacity estimation. To validate the effectiveness of the proposed method, ten batteries are performed for aging tests using a non-constant current charging protocol. The validation results show that the proposed method's average capacity estimation error (mean absolute percentage error) is only 0.65 %. The estimation error is reduced by more than 30 % for the new temperature condition. Compared with the three existing classical methods, the capacity estimation accuracy of the proposed method is improved by 69 %, 82 %, and 68 %, respectively. Additional experiments demonstrate the generality of the proposed approach for different battery chemistries and charging protocols. This work opens the door to battery capacity estimation for non-constant current charging scenarios.

1. Introduction

Due to their excellent performance, lithium-ion batteries have been widely used in electric vehicles, mobile robots, wearable devices, and energy storage stations [1,2]. However, nonlinear and strongly time-varying capacity degradation inevitably occurs during battery usage, which in turn affects battery performance [3,4]. To ensure the safety and reliability of batteries, fast and accurate capacity estimation is essential [5]. Further, accurate capacity estimation helps to achieve accurate state-of-charge (SOC), state-of-energy (SOE), and driving range estimation [6]. Nevertheless, accurate and robust online capacity estimation under real-world conditions is challenging due to variable charging and discharging modes, operating temperatures, and user habits [7,8].

Currently, the flexibility, adaptability, and simplicity of data-driven approaches have made them a prominent methodology in battery capacity estimation [9,10]. Typically, data-driven methods extract health features from the charge or discharge curves of a battery that can characterize battery degradation. Then, a machine learning (ML) model is trained to learn the mapping between these features and battery capacities. Finally, sample data with unknown capacity are estimated using the learned model [11,12]. Compared to the dynamic and variable discharge conditions, the charging process is more stable and facilitates capacity estimation [13]. A large number of data-driven models have been developed for battery capacity estimation when charging, especially under constant current charging scenarios [14].

The most typical method is based on incremental capacity analysis (ICA) [15]. Using ICA, it is possible to convert the ambiguous voltage plateaus on the constant current charging curves of the battery into clearly visible peaks on the incremental capacity (IC) curves [16]. The evolution of these peaks has been proven to be closely related to the degradation mechanisms within the battery, i.e., the loss of active

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materials (LAM) and loss of lithium inventory (LLI) [17,18]. Naturally, it has been extremely effective in extracting features from IC peaks, such as peak height [19], peak position [20], peak slope [21], and peak area [22], and integrating them with ML algorithms to estimate capacity. In addition to voltage information, some methods used local charging capacity to construct health features. Braco et al. [23] directly extracted the charging capacity within different local voltage regions as health indicators. Pearson correlation coefficient (PCC) and linear regression (LR) were used for feature selection and capacity estimation modeling, respectively. Similarly, Huang et al. [24] utilized the genetic algorithm (GA) to determine the combination of charging capacities of specified voltage ranges to estimate the battery capacity. Deng et al. [25] extracted the mean and standard deviation of the charging capacity sequences of different voltage ranges from the constant current charging process as health features. Highly computationally efficient sparse Gaussian process regression (GPR) was used to establish a mapping between features and capacities. Lyu et al. [16] attempted to combine IC peak and charging capacity features to describe battery capacity degradation. Furthermore, to avoid the hassle and unreliability of manual feature design, some recent works have focused on directly inputting raw collected data into ML models for capacity estimation. Tian et al. [26] fed the charge voltage data collected within any 400 s directly into a convolutional neural network (CNN) to achieve the joint estimation of battery capacity and SOC. Similarly, Li et al. [27] used voltage sequences of 3.65 V to 3.89 V from NMC cells as direct inputs to a long short-term memory (LSTM) network to estimate battery capacity. Unsurprisingly, direct input methods based on the combination of voltage sequences, capacity sequences, and temperature sequences in constant current charging mode [28–30] also showed satisfactory capacity estimation performance.

Although the aforementioned methods have all demonstrated the effectiveness of data-driven methods for battery capacity estimation, they all share one underlying assumption: the charging process is constant current charging. However, data-driven methods should be adapted to different charging protocols [31]. Noteworthy, non-constant current charging is an important and non-negligible charging mode for practical applications. Unfortunately, existing methods may deteriorate in non-constant current charging scenarios. For example, the mapping between IC curve (derived from voltage response) and battery aging will be altered by the non-constant current excitation. The regional charging capacity features corresponding to the IC peak area will change. In fact, for constant current charging, the work of Lin et al. [32] showed an equivalent relationship between constant current charging time, charging capacity, and IC peak area. Nevertheless, this equivalence relationship will be broken due to non-constant current charging conditions. Obviously, some time-based health features [15] will lose their physical meaning and become invalid. Therefore, developing data-driven capacity estimation models for non-constant current charging is urgent and unexplored.

Another requirement for online capacity estimation methods is rapidity and low computational complexity. Although some works have focused on the training time (second level) [33] or the inference time (millisecond level) [34] of data-driven models, they are not in the same order of magnitude as the feature collection time (minute level). From this perspective, the feature collection time directly dominates the capacity estimation process, which in turn affects the user's waiting time. Most existing segment data based methods rely on a voltage range of at least 0.2 V to capture IC peak-related features [19,35] or design region capacity features [25]. A voltage window of 0.2 V near the IC peak requires approximately at least 20 min for 1 C constant current charging, which is time-consuming. On the other hand, previous studies have typically used multiple highly correlated health features to estimate capacity [21,24,27]. The large feature input dimension not only increases the burden of hardware acquisition, but also easily falls into the dilemma of multicollinearity [6,36]. In addition, complex feature preprocessing, such as filtering [35] and numerical integration [24], further increases the concern about the practicality of these methods. Therefore, it is crucial to design short-term and low-complexity health features, especially for non-constant current charging.

Moreover, the generalizability of capacity estimation methods to different temperature environments is a key challenge [8,9,25]. The temperature has been shown to affect battery capacity significantly [23]. Chen et al. [37] found that a model established at 25 °C was directly used for 10 °C low-temperature with large errors in capacity estimation. Nevertheless, there are few studies on transfer estimation for variable temperature conditions [38]. Several existing studies have attempted to estimate battery capacity using surface temperature variations during battery charging, such as averaging surface temperature [39], temperature sequences [40], differential temperature curves [41], and the area under the differential thermal voltammetry (DTV) curve [42]. Unfortunately, these works are still based on constant temperature environments. The effectiveness of the developed models for variable temperature environments needs to be further validated. Some other works [43–46] have demonstrated the adaptability of the proposed models for different operating temperature cases. However, estimating capacity at new temperature conditions requires re-modeling, which essentially does not achieve capacity transfer estimation under variable temperature conditions. Crucially, to avoid time-consuming aging tests in new temperature conditions, it is important to use as little early data (several initial cycles) as possible to achieve capacity transfer estimation. In summary, there is an urgent need to investigate an efficient transfer learning strategy to improve the capacity estimation accuracy of data-driven models under variable temperature conditions.

To bridge the above research gap, a data-driven battery capacity estimation method combining GPR and LR is proposed in this work. It can not only quickly estimate the battery capacity in non-constant current charging scenarios, but also conveniently migrate to batteries operating under other temperature conditions. The main contributions of this article are listed as follows.

(1) A non-constant current charging protocol derived from a real vehicle is utilized for battery cycling aging tests. The corresponding non-constant current battery degradation dataset is reported and investigated for the first time.
(2) A single health feature in the high charging voltage region is captured. This feature can be collected within 3 min, and its correlation with battery capacity is over 0.99. Fast and accurate capacity estimation in non-constant current charging scenarios is achieved by combining a GPR model.
(3) A transfer strategy based on the linear transformation of a small amount of early data is proposed to achieve transfer estimation under variable temperature. Validation results demonstrate that the proposed method significantly improves capacity estimation performance under both low and high temperature conditions.
(4) A systematic comparison demonstrates the great advantages of the proposed method over classical methods in terms of estimation time, estimation accuracy, computational complexity and practicality. Moreover, additional experiments show the generality of the proposed method across different battery chemistries and charging protocols.

The remainder of this article is organized as follows. Section 2 presents the battery aging experiments, the selected health features, the machine learning algorithms, and the details of the transfer learning strategy. Section 3 analyzes and discusses the experimental verification results. Finally, conclusions are given in Section 4.

2. Data and methodology

A fast and transferable data-driven approach is proposed for battery capacity estimation in non-constant current charging and variable temperature scenarios, as shown in Fig. 1. In the offline training stage,

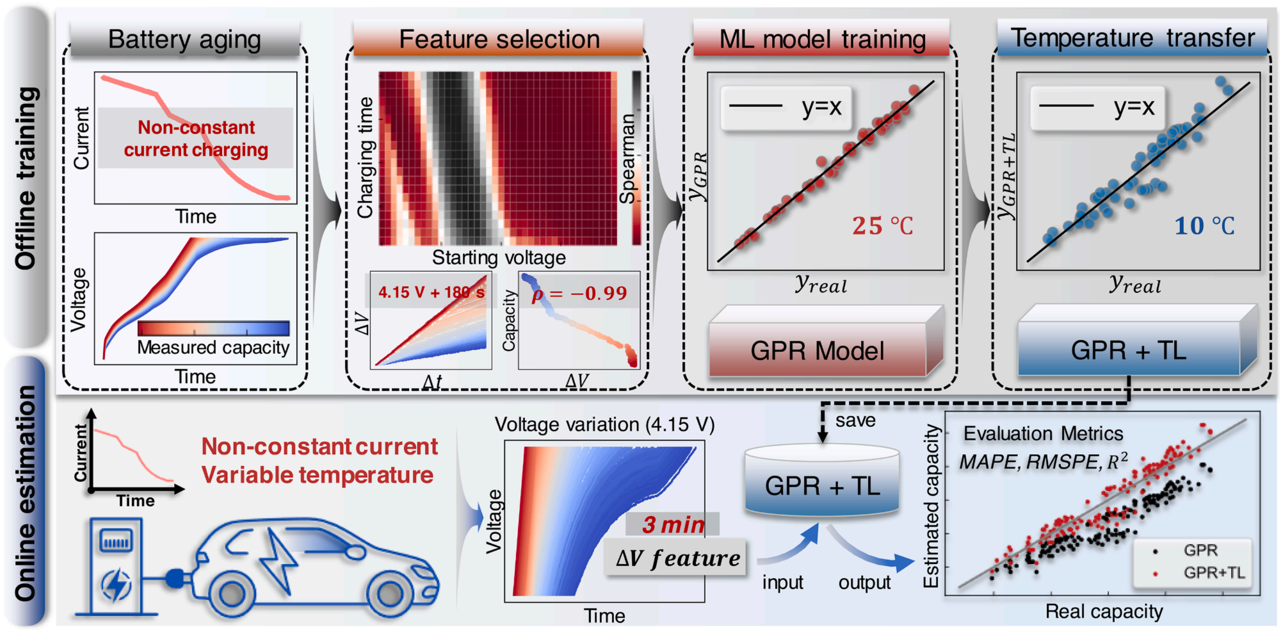

Fig. 1. Overview of data-driven battery capacity estimation under non-constant current charging and variable temperature.

battery aging experiments based on non-constant current charging are performed to collect capacity degradation data. Correlation analysis is used for health feature selection. The health features collected within 3 min in the high charging voltage region (>4.15 V) are utilized to characterize the battery degradation. Then, a GPR model is trained to learn the mapping between features and battery capacities (at 25 °C). Under the new temperature conditions, variable temperature transfer estimation is achieved using several early cycles data combined with linear transformation-based transfer learning (TL). The already trained GPR+TL model is saved. For the online estimation scenario, the voltage response data of the power battery in non-constant current charging mode is recorded. The voltage incremental feature collected within 3 min is extracted and used as input to the GPR model. The current ambient temperature is utilized to determine whether to apply the transfer learning strategy. Finally, the GPR+TL model directly outputs the online capacity estimation results.

2.1. Battery aging experiments with non-constant current charging

Eight commercial 18,650 cells with $LiCoO_2$ positive electrode (LCO battery) were used for aging tests in this study. Their nominal capacity is 2.55 Ah. Four of them (Cells 1–4) were used for normal temperature aging (25 °C). Cells 5–6 and Cells 7–8 were used for low-temperature (10 °C) and high-temperature (40 °C) aging, respectively. In contrast to the common constant-current-constant-voltage (CCCV) charging mode, a four-stage variable current charging mode was used for the charging process of these cells, as shown in the charging current profile in Fig. 2(a). This non-constant current charging mode was derived from a real charging scenario of an electric vehicle, details of which can be found in supplementary note 1. The constant 2 C-rate (5.1 A) current was used to discharge these cells to the lower cut-off voltage (2.75 V), and after a 5 min rest, 0.5 C-rate (1.275 A) current was used to discharge the cells again to 2.75 V. The battery capacity for each cycle can be obtained by integrating the discharge current with time. Fig. 2(a) shows the charge/discharge current profile for a typical cycle and its corresponding voltage response.

Fig. 2(b) shows the capacity degradation trajectories of Cells 1–4 under the constant temperature (25 °C). After at least 300 cycles, the capacity of these cells decays to about 2 Ah. Probably due to the influence of variable current charging, the degradation trajectories of all four cells exhibit drastic fluctuations, which increases the difficulty of online capacity estimation. Fig. 2(c) shows the degradation process of the charging curves of Cell 1. It can be seen that the charging voltage curve gradually shifts to the left and becomes steeper as aging deepens. The aging cell is charged to the upper cut-off voltage (4.2 V) in a faster time. For a clearer view, the reciprocal changes of the slope of the voltage curve (dt / dV) for the first cycle and the last cycle (Cycle=358) are shown in the embedded plot of Fig. 2(c), respectively. It can be observed that the voltage slope increases overall with battery aging (black line to red line) when the voltage is less than about 3.9 V. In contrast, in the local region of [3.9, 4.0] V, the voltage slope decreases with battery aging. More notably, the overall voltage slope increases more significantly with battery aging in the high voltage region (>4.1 V). The voltage slope increase after 4.15 V is more than 300 %. This implies that

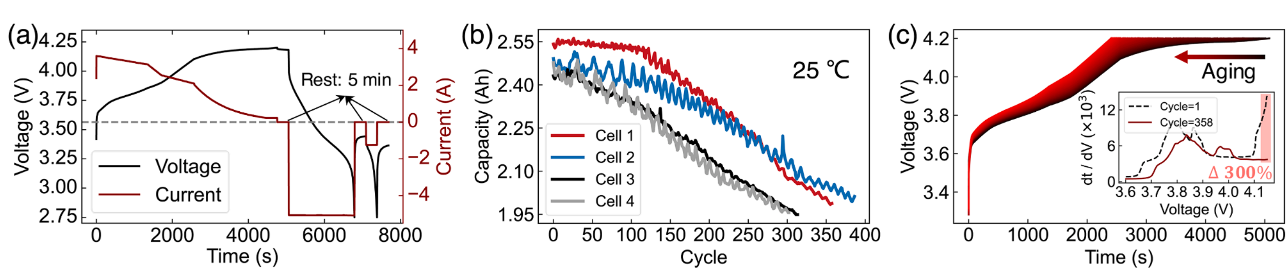

Fig. 2. Battery non-constant current charging aging experiment. (a) Charge/discharge profile of a typical cycle. (b) Capacity degradation trajectories of Cells 1–4. (c) Degraded charging voltage profiles for Cell 1.

the increase in terminal voltage is greater for cells with greater charge voltage slope in the same charging time. The discovery of this region inspired us to use the voltage increment to construct high-quality health features.

2.2. Analysis and selection of health features

Health feature selection is critical for data-driven battery capacity estimation. The quality of the features often determines the model's performance. Cell 1 is chosen as a case study for screening high-quality health features in this section. Based on the observation of the slope of the voltage curve in Fig. 2(c), the voltage increments for different starting voltages over different charge durations are comprehensively investigated.

Fig. 3(a) shows the schematic diagram of feature extraction. For each charging voltage curve during the battery life cycle, the voltage increment (ΔV) could be calculated by charging the same time (Δt) at the same starting voltage. The resulting voltage increment sequence $[\Delta V_1, \Delta V_2, \cdots, \Delta V_n]$ is highly correlated with the capacity sequence $[Q_1, Q_2, \cdots, Q_n]$. For a comprehensive study, the range of charging times and starting voltages was extended. The charging time consisted of segments spaced 30 s apart in the range [30, 600] s. Charging at larger starting voltages will cause the cell to reach the cut-off voltage in a very short time, which may lead to feature failure. Meanwhile, smaller starting voltages (lower SOC) are rarely encountered in the daily charging of electric vehicles. So, the starting voltage was chosen as points spaced at 0.01 V in the range [3.80, 4.15] V. On this basis, 720 (20 × 36) combinations could eventually be formed. Among these combinations, the Spearman correlation coefficients [47] between the voltage increment sequence and battery capacity sequence corresponding to each combination are calculated, as shown in Fig. 3(b). The red gradient color blocks represent the negative correlation between the voltage increment sequence and the capacity sequence, i.e., the voltage increment gradually increases as the battery capacity decreases. On the contrary, all the black gradient color blocks represent positive correlations. It can be observed that the positive correlation color blocks are clustered in the [3.9, 4.0] V region. It indicates that the voltage increment in these regions decreases with the decrease of battery capacity. It is consistent with the decrease in the slope of the voltage curve in the corresponding region in the embedded plot of Fig. 2(c). In addition, the regions where negative correlations appear are within expectations in combination with the analysis in Section 2.1.

Fig. 3(c) shows the highest negative and positive correlations within different charging time segments. Each red square and black circle represents the darkest red and black block in each row of Fig. 3(b), respectively. The detailed numerical results can be found in supplementary note 2. It can be observed that negative correlations are stronger than positive correlations in all charging time segments. For the negative correlation, the voltage increment sequence corresponding to 180 s of charging at 4.15 V achieves the highest correlation coefficient of 0.9933 with the capacity sequence. For the positive correlation, the highest correlation coefficient of 0.9273 occurs at 3.93 V ($\Delta t = 270$s). The visualization results of the above two feature sequences with the capacity sequence are shown in Fig. 3(d) and Fig. 3(e), respectively. The battery voltage variations for the corresponding charging time periods are shown in the embedded plots. The voltage increment sequence corresponding to "4.15 V + 180 s" shows a monotonic phased linear mapping with the battery capacity sequence during the battery life cycle. However, the voltage increment sequence corresponding to "3.93 V + 270 s" shows a non-monotonic and nonlinear relationship with the battery capacity sequence. These results suggest that the feature corresponding to "4.15 V + 180 s" is a better indicator of battery capacity degradation. Therefore, considering the correlation, sampling time, and adaptability, the voltage increment obtained by charging at 4.15 V for 180 s is chosen as the optimal health feature. This feature is used for subsequent machine learning modeling, providing an important basis for achieving fast battery capacity estimation in 3 min. Moreover, only the voltage at the end moment needs to be recorded to obtain the proposed feature, which is very convenient for implementing hardware facilities in practical applications.

In battery usage, two main degradation modes, namely loss of lithium inventory (LLI) and loss of active material (LAM), have unique effects (offset or shrinkage) on the open circuit voltage (OCV) of the cathode and anode, thereby altering the full voltage curve (see Fig. 2(c)) and lowering the total capacity [48]. Hence, the charging curves contain rich information about degradation process (especially in the graphite anode) [49]. The ΔV feature can be regarded as a measure of the local information (for practicality) of the charging curve. It is associated with energy dissipation and internal resistance buildup [50]. Based on this, the optimal ΔV feature can be extracted from numerous local information using correlation analysis to accurately indicate battery degradation. Overall, the selected ΔV feature can be seen as an externally

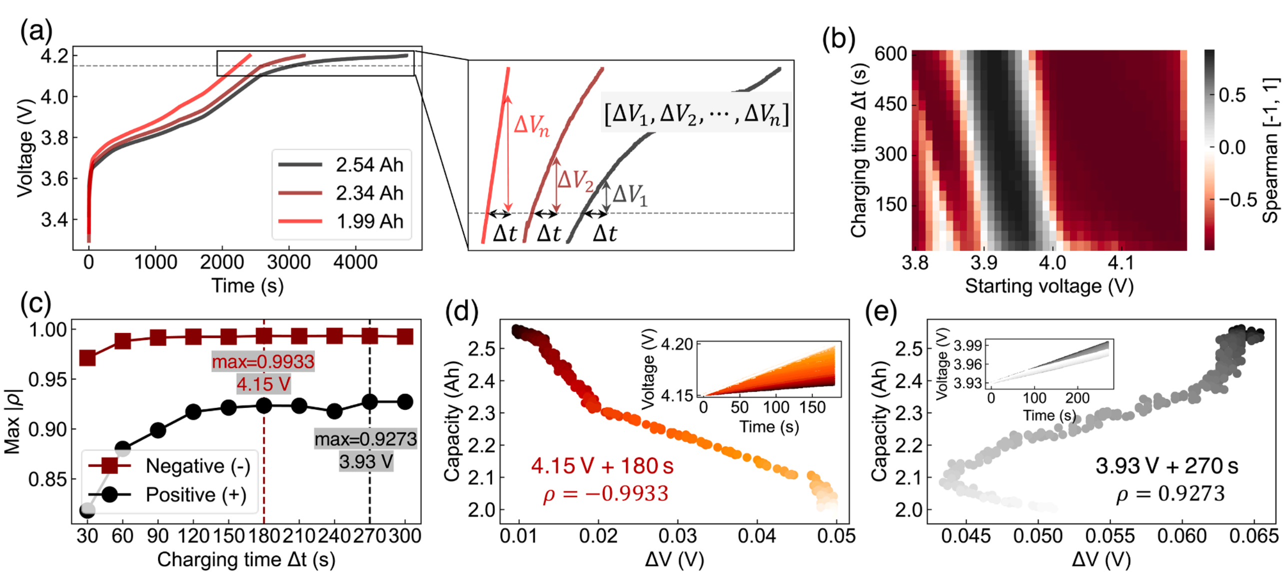

Fig. 3. Battery health feature analysis and selection (Cell 1). (a) Schematic diagram of feature extraction. (b) Spearman correlation heat map of feature sequences and capacity sequences corresponding to different combinations. (c) Highest negative and positive correlation coefficients in different charging time segments. Feature visualization for (d) "4.15 V + 180 s" and (e) "3.93 V + 270 s".

measurable representation of the internal degradation modes of the battery, which provides a physical interpretation for the ΔV feature.

2.3. Machine learning regression models

Considering the simplicity of the proposed health features and the training data size, a non-parametric Gaussian process regression (GPR) model was adopted for modeling. GPR uses a Gaussian process (GP) to handle the data regression problem. A GP can be defined as a collection of finite random variables which follow a Gaussian distribution [51]. The GP is generally denoted as

$$f(x) \sim GP(m(x), \kappa(x, x')) \tag{1}$$

where $m(x)$ and $\kappa(x, x')$ are the mean and covariance functions, respectively, denoted by

$$m(x) = E[f(x)] \tag{2}$$

$$\kappa(x, x') = E\left[(f(x) - m(x))(f(x') - m(x'))^T\right] \tag{3}$$

where $m(x)$ is usually set to 0. $\kappa(x, x')$ is also called the kernel function of the GPR and is used to describe the relationship between two observations x and x'. The choice of kernel function has a significant impact on the performance of GPR. The Matern kernel has been reported to achieve the best performance in several battery capacity estimation tasks [51–53]. It is expressed as follows:

$$\kappa_{Ma}(x - x') = \sigma_f^2 \frac{2^{1-\upsilon}}{\Gamma(\upsilon)} \left(\sqrt{2\upsilon}\ \frac{(x - x')}{\rho}\right)^{\upsilon} R_{\upsilon}\left(\sqrt{2\upsilon}\ \frac{(x - x')}{\rho}\right) \tag{4}$$

where $\Gamma(\upsilon)$ is the gamma function and R_υ is a modified Bessel function. The Matern kernel with smoothness hyperparameter $\upsilon = 3\ /2$ was adopted in this paper.

For a typical regression problem, the observations usually contain Gaussian white noise (with the zero mean and the variance σ_n^2). By assuming that $m(x)$ of GP is 0, the prior distribution of observations can be expressed as follows:

$$y \sim N\left(0, K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I}\right) \tag{5}$$

where $K(\mathbf{X},\ \mathbf{X})$ is the covariance matrix, and the element of the matrix is given by $K_{ij} = \kappa(x_i, x_j)$. Based on the assumption that the training and test sets follow the same Gaussian distribution, for a new test set input x^*, the joint prior distribution between the training set output $\boldsymbol{y}$ and the test set output y^* can be expressed as:

$$\begin{bmatrix} \mathbf{y} \\ y^* \end{bmatrix} = N\left(0, \begin{bmatrix} K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I} & K(\mathbf{X}, x^*) \\ K(x^*, \mathbf{X}) & K(x^*, x^*) \end{bmatrix}\right) \tag{6}$$

Then, according to the Bayesian formula, the posterior distribution of y^* can be calculated as follows:

$$p(y^* | \mathbf{X}, \mathbf{y}, x^*) = N(y^* | m^*, \sigma^*) \tag{7}$$

where

$$m^* = K(x^*, \mathbf{X})\left[K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I}\right]^{-1} \mathbf{y} \tag{8}$$

$$\sigma^* = K(x^*, x^*) - K(x^*, \mathbf{X})\left[K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I}\right]^{-1} K(\mathbf{X}, x^*) \tag{9}$$

The hyperparameters σ_f and ρ of the kernel function and the variance σ_n^2 can be determined by combining the negative log marginal likelihood method with a gradient-based optimizer [51]. After the above steps, the feature sequences $[\Delta V_1,\ \Delta V_2,\ \cdots,\ \Delta V_n]$ are fed into the established GPR model and the corresponding estimated capacity sequences $[Q_1,\ Q_2,\ \cdots,\ Q_n]$ can be obtained.

To demonstrate the superiority of GPR, several prevalent non-probabilistic machine learning models, linear regression (LR) [23], support vector regression (SVR) [41], and random forest (RF) [32] were chosen as benchmarks. In the present study, the above mentioned models were implemented using the sklearn package (version 1.0.2) of Python 3.7.6. The hyperparameters of SVR and RF were determined by 5 times 5-fold random search (taking the best one) on the training set.

2.4. Linear transformation based transfer learning

When the external temperature conditions change, the capacity estimation error of the data-driven model will increase. To improve the estimation accuracy of the model under different operating temperatures, a transfer learning (TL) strategy based on linear transformation of early data is developed in this paper, as shown in Fig. 4. When the ΔV features collected under other temperature conditions are directly input to the established GPR model (25 °C), a large estimation bias will occur. To reduce this temperature-induced bias, an extrapolatable LR model is added after the output layer of the GPR model. LR uses the early aging data to establish a mapping relationship between the GPR output and the true value, which can be expressed as:

$$Q' = k \cdot Q + b \tag{10}$$

where Q and Q' represent the GPR and LR outputs, corresponding to y_{GPR} and y_{GPR+LR} in Fig. 4, respectively; k and b denote the slope and intercept of the LR model, respectively.

The above strategy requires data from only the first few cycles of the variable temperature aging experiment to achieve transfer learning (reducing the estimation error), which maximally ensures the practicality of this TL strategy. The exact number of early cycles required for transfer learning is further discussed in Section 3.2. In summary, after two levels of prediction by GPR and LR, the prediction bias caused by temperature will be suppressed. On this basis, the proposed method is able to achieve better capacity estimation under variable temperature conditions. More importantly, due to the simplicity of the proposed TL strategy, the method can be easily extended to other temperature conditions. This provides the basis for efficient modeling of capacity estimation in the entire temperature domain of the battery.

2.5. Model evaluation criteria

In the present paper, considering the different absolute cell capacities, four metrics: the percentage error (PE), the mean absolute percentage error (MAPE), the root mean square percentage error (RMSPE), and the model determination coefficient R^2 are used to measure model performance. Their computational expressions are as follows:

$$PE = \frac{\widehat{y}_i - y_i}{y_i} \times 100\% \tag{11}$$

$$MAPE = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{\widehat{y}_i - y_i}{y_i} \right| \tag{12}$$

$$RMSPE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \left(\frac{\widehat{y}_i - y_i}{y_i} \right)^2} \tag{13}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (\widehat{y}_i - y_i)^2}{\sum_{i=1}^{n} (y_i - \overline{y})^2} \tag{14}$$

where y, $\widehat{y}$, and $\overline{y}$ represent the real value, predicted value, and average value of battery capacity, respectively. n is the total number of test samples. The closer PE is to 0 indicates smaller error. Similarly, smaller MAPE and RMSPE indicate higher model estimation accuracy. The closer R^2 is to 1 indicates better model prediction performance.

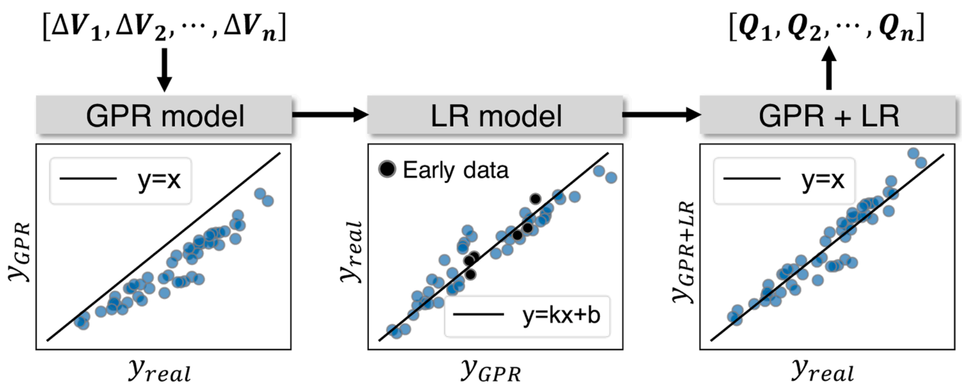

Fig. 4. Schematic diagram of a transfer learning strategy based on linear transformation of early data.

3. Results analysis and discussions

In this section, first, the validation results of the extracted single ΔV feature (see Section 2.2) combined with ML models at 25 °C are reported (Cell 1 used for training, Cells 2–4 used for testing). The comparison of LR, GPR, SVR, and RF shows that GPR is the optimal estimator. In Section 3.2, the GPR model (established at 25 °C) combined with the proposed transfer method (see Section 2.4) is applied to 10 °C (Cells 5–6) and 40 °C conditions (Cells 7–8) to validate the performance under varying temperatures. In Section 3.3, the proposed method is compared with existing typical data-driven methods, demonstrating its advantages for non-constant current charging scenarios. Section 3.4 additionally validates the performance of the proposed method on 2 NCM cells. Finally, the adaptability to constant current charging protocols (using 8 LCO/NCO cells) and future work are discussed in Section 3.5.

3.1. Capacity estimation results

Without loss of generality, Cell 1 was chosen as the training set, and Cells 2–4 were chosen as the test set. The capacity estimation errors of the four ML models on Cells 2–4 are shown in Fig. 5(a)–(c). The maximum estimation error of the four ML models is almost within 0.1 Ah for a cell with a rated capacity of 2.55 Ah, which proves the validity of the extracted health feature (ΔV). As expected, LR has the worst performance due to its inability to capture the nonlinear mapping relationship between ΔV features and capacities (see Fig. 3(d)). However, the remaining three nonlinear ML models outperform LR significantly. Specifically, the boxplot of GPR displays the flattest main body, showcasing the optimal performance and indicating that errors smaller than 0.02 Ah are observed in over 75 % of the samples. In comparison, the performance of SVR and RF is slightly inferior. Table 1 shows the detailed numerical results. The average MAPE and RMSPE of GPR on the three tested cells are 0.65 % and 0.99 %, respectively. From the perspective of average error, GPR achieves the lowest estimation error by a narrow margin. Meanwhile, GPR gets the highest average R^2 score of 0.9830. In addition, compared to SVR and RF, GPR does not require a tedious and time-consuming parameter adjustment process. Therefore, considering the estimation accuracy and computational efficiency, GPR is identified as the optimal estimator. The capacity estimation results of GPR on Cells 2–4 are shown in Fig. 5(d)–(f), respectively. It can be observed that the estimated capacities are very close to the real capacities. Specifically, the MAPEs of all three tested cells are less than 0.8 %, and the estimation error is within 2 % for most samples. The R^2 of all three tested cells is greater than 0.98. These results indicate the superior estimation performance of GPR. Moreover, it only takes 3 min to complete the feature sampling to output the capacity estimation results

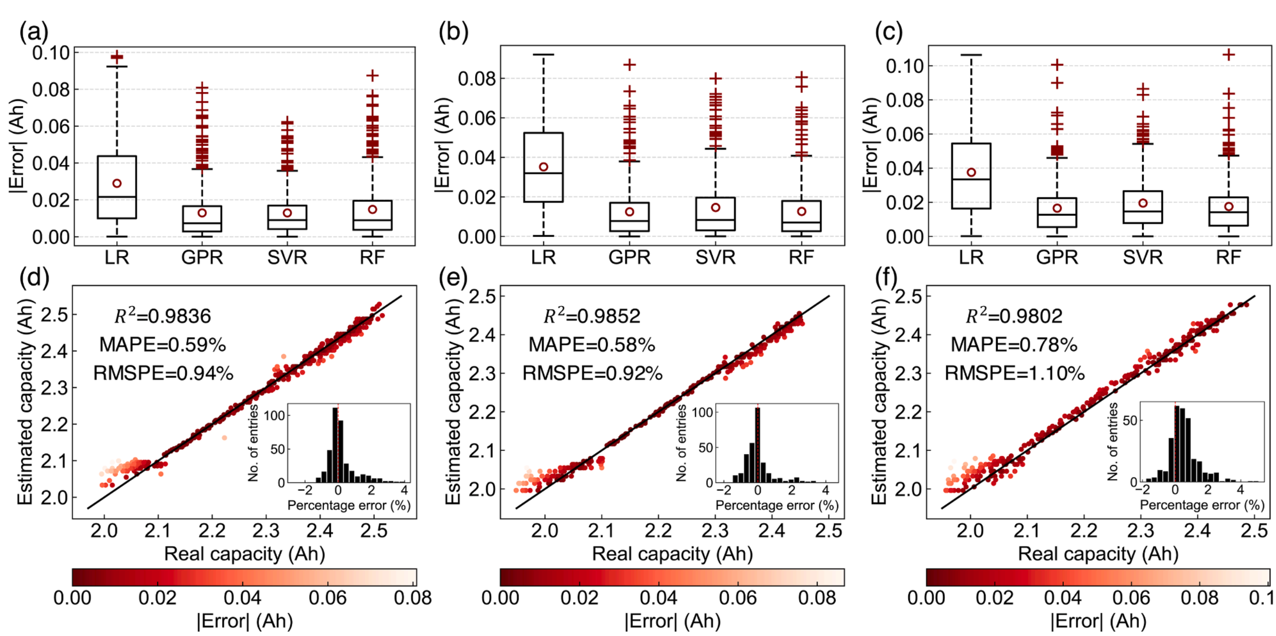

Fig. 5. Capacity estimation error boxplots for (a) Cell 2, (b) Cell 3 and (c) Cell 4 (red circles represent the mean of absolute errors). Capacity estimation results of GPR on (d) Cell 2, (e) Cell 3 and (f) Cell 4.

Table 1
Comparison of capacity estimation errors (%) and R^2 for different ML models.

Cells No.	LR			GPR			SVR			RF		
	MAPE	RMSPE	R^2	MAPE	RMSPE	R^2	MAPE	RMSPE	R^2	MAPE	RMSPE	R^2
Cell 2	1.27	1.61	0.9420	0.59	0.94	0.9836	0.58	0.82	0.9865	0.68	1.05	0.9792
Cell 3	1.58	1.84	0.9298	0.58	0.92	0.9852	0.69	1.09	0.9796	0.59	0.94	0.9845
Cell 4	1.68	2.02	0.9194	0.78	1.10	0.9802	0.91	1.21	0.9750	0.82	1.13	0.9787
Mean	1.51	1.82	0.9304	**0.65**	**0.99**	**0.9830**	0.73	1.04	0.9804	0.70	1.04	0.9808

during each non-constant current charging process, which is very beneficial to the practical deployment of the model. Furthermore, the results of leave-one-out cross-validation [41] (see supplementary note 3) further validate the generalization of the proposed method. In short, the above analysis demonstrates that the proposed data-driven approach can achieve accurate and efficient online battery capacity estimation in non-constant current charging scenarios.

3.2. Temperature transfer estimation results

To verify the effectiveness of the proposed TL strategy under variable temperature conditions, Cell 5 and Cell 6 were subjected to aging tests in a 10 °C environment. Except for the ambient temperature, their other test conditions remain consistent with Cells 1–4. Fig. 6(a) shows the aging trajectories of Cell 5 and Cell 6. Under the 10 °C condition, it can be observed that the initial capacity of the cells is only about 2.1 Ah. The whole trajectories show a trend of increasing and then decreasing. After about 140 cycles, the capacity of both cells drops below 80 % (2.04 Ah) of the rated capacity. Compared to Cells 1–4 (25 °C, see Fig. 2(b)), the low temperature of 10 °C increased the aging rate by roughly 60 %. Worse still, the uncertainty fluctuations in capacity became more dramatic. These phenomena may be related to the polarization increase and lithium plating of graphite negative electrodes [54]. It is conceivable that the GPR model built at 25 °C is directly used for capacity estimation in a 10 °C environment with large errors. Therefore, improving the estimation accuracy using the transfer learning strategy is urgent.

The first 6 cycles of Cell 5 and Cell 6 were used for the transfer learning modeling (see Section 2.4). It is worth noting that the 6 cycles only occupied about 4 % of the full life-cycle data, which is a very small sample transfer size. For the rest of the tested cycles, the estimation results of the GPR direct estimation (GPR) and GPR combined with TL strategy (GPR+TL) on Cell 5 and Cell 6 are shown in Fig. 6(b) and (c), respectively. It can be observed that the introduction of the TL strategy makes the estimated capacity closer to the real capacity. In particular, for Cell 5, without using TL, the capacities would be severely underestimated, with a maximum error of more than 6 %. Using TL, the

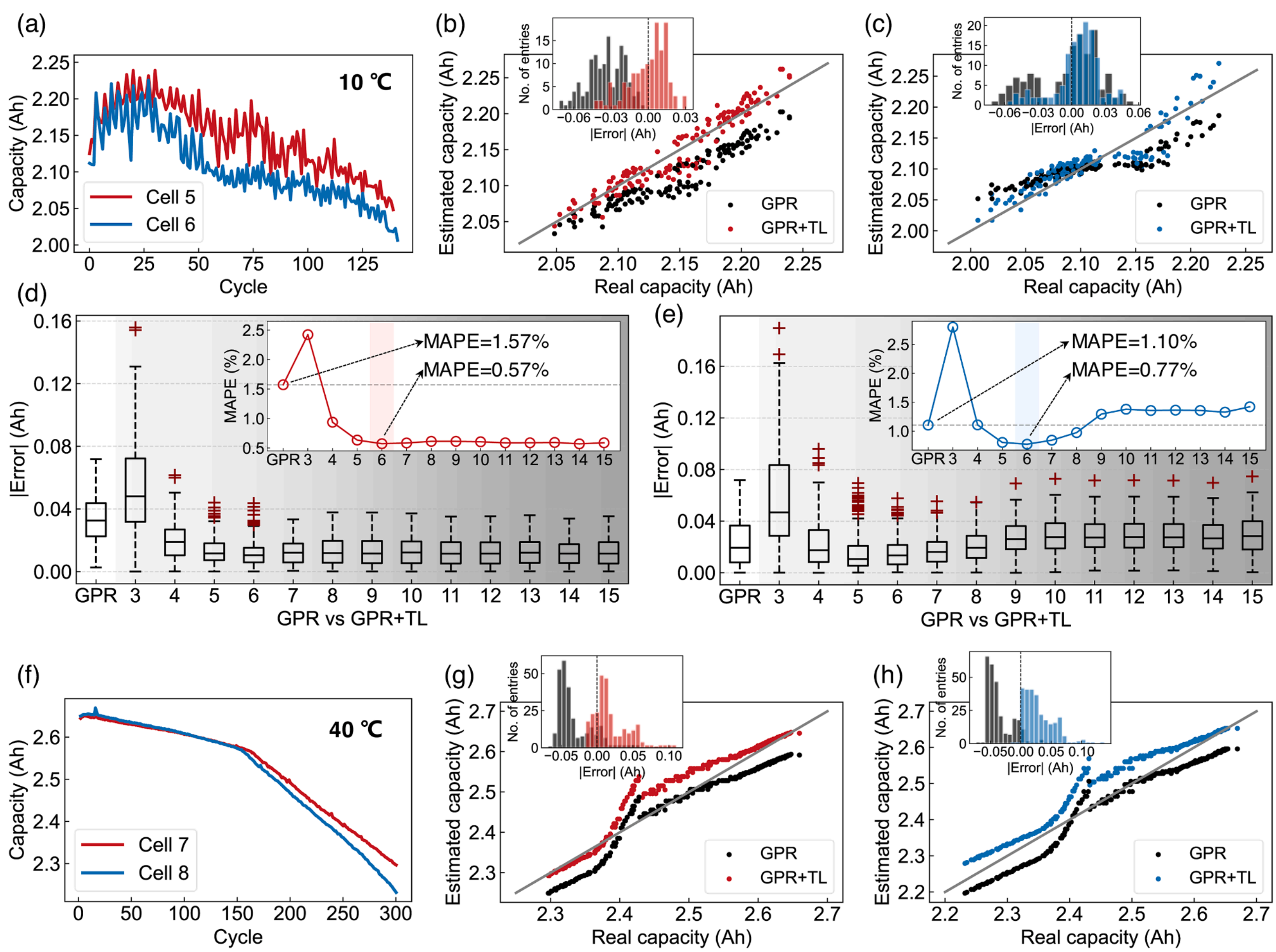

Fig. 6. Temperature transfer estimation results. Battery degradation trajectories at (a) 10 °C and (f) 40 °C. Capacity estimation results of GPR and GPR+TL methods on (b) Cell 5, (c) Cell 6, (g) Cell 7, and (h) Cell 8. The effect of different early cycle numbers on transfer learning for (d) Cell 5 and (e) Cell 6.

estimation errors are corrected to within the range of ±3%. Table 2 shows the detailed numerical comparison results. The GPR model without TL strategy is used as a zero-shot learning (ZSL) baseline. Also, the GPR model is retrained using equal early cycle data from Cell 5 or Cell 6 as a No TL comparison. By incorporating the TL strategy, the capacity estimation error (MAPE) is reduced by 63 % and 30 % for Cell 5 and Cell 6, respectively. The R^2 for Cell 5 is improved from 0.4173 to 0.8962, and the R^2 for Cell 6 is improved from 0.6545 to 0.8235. This is a huge performance improvement. In contrast, No TL obtains extremely poor results with R^2 less than 0. This indicates that the GPR model trained using only 6 cycles cannot accurately estimate the future battery capacity.

To investigate the effect of migrating sample data volume on the TL strategy, different numbers of early cycles were used for linear transformation-based TL modeling. For a fair comparison, all cycles of the cells were used as the test set. The estimation errors of the different methods for Cell 5 and Cell 6 are shown in Fig. 6(d) and (e), respectively. For Cell 5, using only the first 4 cycles for transfer learning reduces the estimation error (MAPE) from 1.57 % (ZSL) to 0.94 %. Using the first 6 cycles for transfer learning reduces the estimation error to a minimum of 0.57 %. Introducing more cycles for transfer learning, there is no significant performance improvement. For Cell 6, transfer learning using the first 6 cycles reduces the estimation error (MAPE) from 1.10 % (ZSL) to a minimum of 0.77 %. Moreover, although 3 sample points are sufficient for LR modeling, the effect is extremely unstable and even larger than the estimation error of the ZSL. Consequently, in practical applications, at least 4 or more early cycles are guaranteed to be used for transfer learning to achieve performance improvement.

To further validate the performance of the TL method in high-temperature environment, similar 300 cycle aging tests at 40 °C were conducted on Cell 7 and Cell 8. Fig. 6(f) shows their degradation trajectories. Unlike other temperature conditions, the battery degradation do not exhibit significant fluctuations. Furthermore, the capacity during the first 150 cycles exceed 2.57 Ah, even surpassing the maximum capacity of Cell 1 (25 °C environment, see Fig. 2(b)) used for modeling. The increase in apparent capacity at high temperatures is consistent with observations in existing studies [9,25,55]. However, due to these samples falling outside the observation range of the training set, estimating the capacity becomes significantly more challenging, which is a well-known issue in data-driven methods. Similarly, the first 10 cycles of Cell 7 and Cell 8 (3.3 % of the total samples) were used for transfer learning modeling, while the remaining 290 cycles served as testing samples. Fig. 6(g) and (h) illustrate the results of GPR (ZSL baseline) and GPR+TL, with numerical results shown in Table 2. As expected, for ZSL, samples outside the training set (capacity greater than 2.57 Ah) are severely underestimated. The same applied to samples with a true capacity less than 2.4 Ah. In contrast, using transfer learning significantly improves this situation. Specifically, through TL, the MAPE of Cell 7 and Cell 8 decreases from the original 1.44 % and 1.51 % to 0.9 % and 1.23 %, respectively. The estimation errors decrease by 38 % and 19 %, highlighting the effectiveness of TL. Furthermore, it can be observed that the extremely limited amount of data is insufficient for GPR to function properly. The R^2 of the No TL baseline remains less than 0, which is unacceptable.

In summary, although the ZSL approach yields decent results, the performance improvement achieved by using a small amount of early data for transfer learning is considerable. The aforementioned analysis demonstrates the potential of the proposed TL method in improving battery capacity estimation under variable temperature conditions. It provides a foundation for rapid capacity estimation modeling of the entire temperature domain of the battery.

3.3. *Comparison with typical methods*

To further verify the superiority of the proposed method, three representative data-driven methods are used for comparison. M1: Incremental capacity (IC) features based method [19,20]. M2: Regional capacity statistical features based method [25,56]. M3: Direct input of charging curves based method [27,57]. Note that all these classical methods are designed for constant-current charging mode, so their validity for non-constant-current charging is an interesting and valuable topic. For a fair comparison, the ML models of all methods are chosen as GPR for demonstration purposes.

Although the incremental capacity analysis (ICA) [15] is proposed for the constant-current charging mode, a similar treatment is applied to the present non-constant-current charging profile. Fig. 7(a) shows the IC curves for the charging process of Cell 1. It can be observed that there is a correspondence between the shift of these curves and battery capacity degradation. Two peaks appear in the regions around [3.78, 3.84] V and [3.86, 3.96] V, respectively, gradually moving down to the right as battery aging. However, unlike the conventional ICA, the evolution of these peaks may not characterize the intrinsic degradation pattern of the battery, such as the loss of active materials and loss of lithium inventory [17], because the constant current charging condition is broken. More importantly, the location and height of these peaks appearing will be affected by the changing charging current. Therefore, the health features based on IC peaks may be unstable. On the other hand, the cumulative charging capacity versus voltage and voltage versus time of the battery at [3.7, 4.0] V are presented in Fig. 7(b) and (c), respectively. Similarly, it can be seen that the evolution of these curves could also reflect battery capacity degradation.

On this basis, the heights and positions of the two IC peaks in Fig. 7 (a) are extracted as health features, as shown in Fig. 7(d)–(g). The Spearman coefficients indicate a high correlation between these features and the battery capacity. The feature with the highest correlation is the peak 2 position, and the feature with the lowest correlation is the peak 1 position. These IC-based features combined with GPR are used as an example of the M1 method. For the ΔQ curve in Fig. 7(b), the mean, standard deviation, and maximum values of ΔQ as well as the mean value of ΔV are extracted as health features as shown in Fig. 7(h)–(k). Note that these statistical features are calculated from the raw sampling points in the time domain. It can be observed that the three statistical features of ΔQ show a similar positive correlation with the battery capacity, while the mean value of ΔV shows a strong negative correlation with the battery capacity. These ΔQ-based features combined with GPR are used as an example of the M2 method. Notably, the correlations of all the above features with capacity are weaker than the proposed ΔV feature (see Fig. 3(d)). Finally, each voltage curve in Fig. 7(c) are linearly interpolated to 100 sample points and used directly as input to the GPR as an example of the M3 method. This approach uses ML models to mine patterns characterizing battery degradation directly from raw data, and it is becoming increasingly popular.

Table 2
Comparison of capacity estimation results for different methods.

Cells No.	GPR (ZSL)			No TL			GPR+TL		
	MAPE	RMSPE	R^2	MAPE	RMSPE	R^2	MAPE	RMSPE	R^2
Cell 5	1.57	1.70	0.4137	5.26	7.00	−8.73	**0.58**	**0.72**	**0.8962**
Cell 6	1.10	1.37	0.6545	8.96	11.43	−21.23	**0.77**	**0.98**	**0.8235**
Cell 7	1.44	1.60	0.8623	3.38	4.67	−0.05	**0.90**	**1.30**	**0.9131**
Cell 8	1.51	1.67	0.8898	39.76	50.13	−87.81	**1.23**	**1.60**	**0.9047**

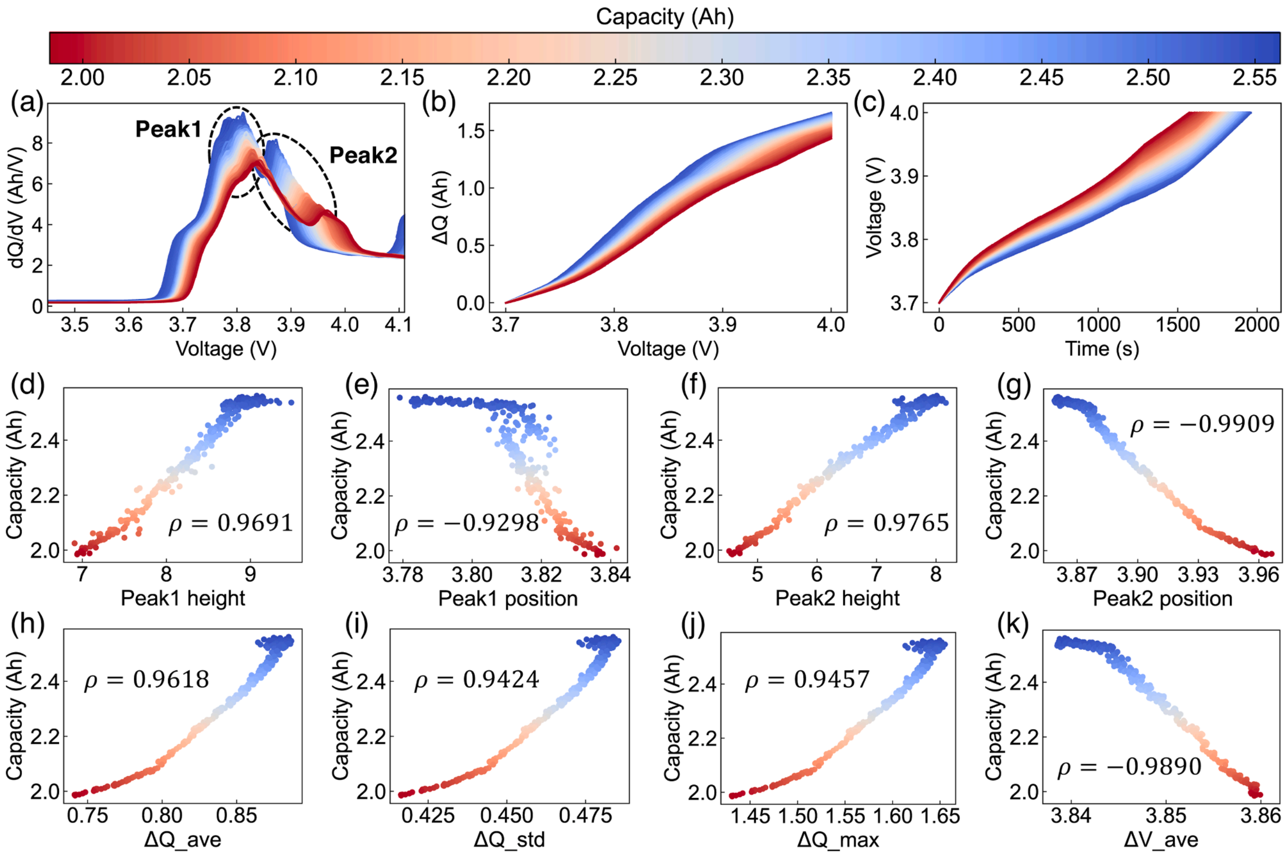

Fig. 7. The feature extraction process of the classical methods (Cell 1). (a) Incremental capacity curves. (b) Cumulative charging capacity versus voltage in the region of [3.7, 4.0] V. (c) Voltage versus time in the region of [3.7, 4.0] V. (d)–(g) Relationships between battery capacity and IC peaks based features. (h)–(k) Relationships between battery capacity and statistical features of regional charging capacity.

Without loss of generality, Cell 1 is used as the training set. The capacity estimation errors of the aforementioned three methods with the proposed method on Cells 2–4 are presented in Table 3. It is worth noting that during the experiments, we found that for M2 and M3, GPR using Matern kernel exhibited poor estimation performance. To obtain more competitive results, the combination of DotProduct kernel and WhiteKernel was used as the kernel function for GPR in M2 and M3. The performance comparison of different kernel functions can be found in supplementary note 4. Table 3 shows that the average MAPE of the three typical methods on all three tested cells exceeds 2 %, which indicates that these methods are not applicable to non-constant current charging scenarios. However, the proposed method performs highest on all three tested cells. Its average estimation error (MAPE) is only 0.65 %, and the average R^2 is 0.9830. Compared to M1, M2 and M3, the average estimation accuracy (MAPE) of the proposed method is improved by 69 %, 82%, and 68 %, respectively. These results show that the proposed health feature ΔV has significant advantages and can be well used to indicate battery capacity degradation during non-constant current charging.

In addition to the estimation accuracy, the proposed method has additional advantages in terms of feature processing. Table 4 shows the feature construction requirements of the different methods. The proposed method requires only one feature collected in 3 min to achieve high-accuracy capacity estimation without any data preprocessing. However, the three typical methods not only require a larger number of features, but also feature sampling times exceeding 26 min. Moreover, some complex data processing operations, such as data differential, integration, and filtering, are required to be applied in the feature construction process. The data processing that determines the model performance greatly increases the computational burden of the BMS hardware facilities and reduces the effectiveness of these typical

Table 4
Feature construction requirements for different methods.

	M1	M2	M3	**Ours**
Number of features	4	4	100	1
Sampling time	>26 min	>26 min	>26 min	3 min
Data processing	Differential, Filtering	Numerical integration	Data interpolation	/

Table 3
Comparison of capacity estimation errors of different methods.

Cells No.	MAPE (%)				RMSPE (%)				R^2			
	M1	M2	M3	Ours	M1	M2	M3	Ours	M1	M2	M3	Ours
Cell 2	1.99	2.87	2.63	**0.59**	2.65	3.16	3.19	**0.94**	0.8318	0.7826	0.7994	**0.9836**
Cell 3	1.83	4.04	1.41	**0.58**	3.49	4.33	1.63	**0.92**	0.7149	0.6247	0.9427	**0.9852**
Cell 4	2.42	4.04	1.99	**0.78**	3.23	4.33	2.20	**1.10**	0.8070	0.6393	0.9036	**0.9802**
Mean	2.08	3.65	2.01	**0.65**	3.12	3.94	2.34	**0.99**	0.7846	0.6822	0.8819	**0.9830**

methods. On the other hand, the charging SOC versus voltage over the full battery life cycle is shown in Fig. 8. Since the voltage range required for the three typical methods is [3.7, 4.0] V, they correspond to a charging SOC range of about 5 %−75 %. The proposed method collects ΔV features within 3 min after charging to 4.15 V (90 % SOC). In general, electric vehicle users rarely discharge their batteries to 20 % SOC or even lower SOC levels [24]. It means that the initial charging SOC of the battery may already exceed the minimum SOC (5 %) required by the three typical methods mentioned above. This further exacerbates the concern about the practicality of these methods. In contrast, since users are more inclined to fully charge the battery in most cases, the range of charging SOC required by the proposed method (>90 %) can be easily satisfied, ensuring practicality. In summary, the above analysis demonstrates the significant advantages of the proposed method in terms of estimation accuracy, estimation time, processing requirements, and practicality. For online capacity estimation of battery non-constant current charging scenarios, the traditional representative methods seem to have failed. There is an urgent need to break this dilemma using the proposed method.

3.4. Validation on NCM cells

To further validate the feasibility of the proposed rapid capacity estimation method for other batteries, two batteries with $LiNiMnCoO_2$ (NCM) cathode were used in aging experiments under a 25 °C environment. The two batteries were dismantled from an accident vehicle and had a nominal capacity of 100 Ah. Similar non-constant current charging mode were adopted (see supplementary note 1). Fig. 9(a) illustrates their degradation profiles. Over the course of more than 40 days, the battery capacity decreased from 99 Ah to 97.5 Ah, showing a relatively linear decline. An experimental interruption at the 100th cycle (approximately 2 days) resulted in slight capacity regeneration. Due to changes in battery chemistry, health features needed to be re-screened. The feature extraction method described in Section 2.2 was performed again, and the results are shown in Fig. 9(b). The features corresponding to most of the regions show a strong negative correlation with battery capacity. The optimal feature was identified as the voltage increment formed after charging 120 s at 4.06 V ("4.06 V + 120 s"). Fig. 9(c) illustrates the comparison between actual capacity and features. The features of both cells show the highest correlation with capacity, reaching −0.9964 and −0.9957, respectively. Using leave-one-out cross-validation, the capacity estimation results of GPR are visualized in Fig. 9 (d) and (e). It can be observed that the estimated values closely matched the true values for 200 cycles. Fig. 9(f) shows the estimation performance indicators. The MAPE for both cells is less than 0.05 %. The R^2 is greater than 0.98. Importantly, the feature collection process could be completed in just 2 min. These results demonstrate the accuracy and rapidity of the proposed method, showcasing its applicability to batteries with other cathode materials.

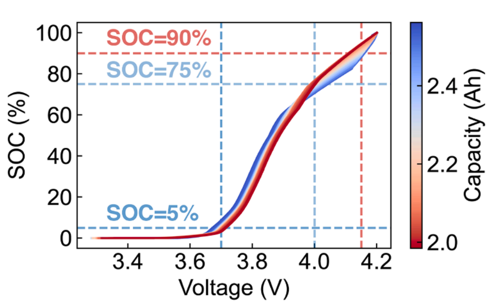

Fig. 8. The relationship between charging SOC and voltage during the whole life cycle of the battery (Cell 1).

3.5. Further discussions

The above results demonstrate the significant advantages of the proposed method in non-constant current charging scenarios. Although capacity estimation during non-constant current charging was the initial focus of this work, the proposed method is equally effective for the constant-current charging mode. Without loss of generality, the widely used Oxford dataset [51] is employed for validation. It consists of eight pouch lithium-ion batteries, referred to as Oxford cells 1–8, with a nominal capacity of 0.74 Ah. The cathode material of these batteries is a blend of lithium cobalt oxide and lithium nickel cobalt oxide (LCO/NCO). These batteries were subjected to a 1 C-rate constant-current charging protocol and operated at a 40 °C environment. Oxford cell 1 was used for feature extraction as described in Section 2.2. Ultimately, the voltage increment after 180 s at 3.79 V ("3.79 V + 180 s") was determined as the most suitable health feature (Spearman correlation coefficient is −0.9991). Based on this, leave-one-out cross-validation [41] was performed on the eight cells. Fig. 10(a) shows the capacity estimation errors of the GPR model. The average MAPE and RMSPE for the eight cells are 0.72 % and 0.97 %, respectively. This is a highly competitive result compared to existing works [26,30,32,41,46], particularly considering the duration of feature collection and computational complexity.

Constrained by the limited scale of the non-constant current charging based battery dataset, to validate the effectiveness of the proposed temperature transfer method more extensively, 55 publicly available NCM batteries from Zhu et al. [9] were used as case studies. These batteries are labeled as NCM_25 cells 1–23 (25 °C), NCM_35 cells 1–4 (35 °C), and NCM_45 cells 1–28 (45 °C) based on their operating temperatures. The rated capacity of these batteries is 3.5 Ah, and they were all charged using the CCCV charging protocol at a rate of 0.5 C and discharged using the CC protocol at a rate of 1 C. Fig. S2(a) shows their aging trajectories (see supplementary note 5). Without loss of generality, NCM_25 cell 1 was used for feature extraction as described in Section 2.2. As shown in Fig. S2(b), the voltage increment after 360 s at 3.52 V ("3.52 V + 360 s") was selected as the optimal feature (Spearman correlation coefficient is −0.99996). Fig. S2(c) visualizes the relationship between the ΔV feature and battery capacity. It can be observed that the ΔV feature of NCM_25 cells 1–23 exhibits a highly linear correlation with capacity. However, there is a significant feature shift between different temperatures, indicating that direct estimation across temperature domains would result in large errors. Fig. 10(b) shows the leave-one-out cross-validation based capacity estimation results for the 23 cells at 25 °C (using GPR). The average MAPE and RMSPE are 0.42 % and 0.46 %, respectively, confirming the effectiveness of the proposed health feature. The GPR model constructed at 25 °C is then applied to other temperature conditions. The capacity estimation errors are shown in Fig. 10(c)-(f). Furthermore, GPR combined with the proposed temperature transfer strategy (GPR+TL) is also used for comparison. For NCM_35 cells 1–4, the first 15 cycles of the 4 cells are utilized for transfer learning. Similarly, for NCM_45 cells 1–28, only the first 15 cycles of cells 1–4 are used for transfer learning to better simulate the real case of variable temperature applications. The effect of the number of transfer samples on performance is presented in Fig. S3 (see supplementary note 6). Note that the samples used for transfer modeling are excluded from the error computation to avoid data leakage. As expected, when GPR is directly applied to the 35 °C and 45 °C conditions, the average MAPE deteriorates to 2.73 % and 4.74 % respectively. It can be observed that the larger the temperature difference, the larger the error. In contrast, by combining the proposed transfer strategy, the capacity estimation errors for all 32 tested cells are reduced. The average MAPE is reduced to 1.81 % and 1.99 %, respectively, indicating an improvement in capacity estimation accuracy by 34 % and 58 %. RMSPE also exhibits substantial improvement. These results demonstrate the significant value of using the proposed temperature transfer strategy to improve the estimation accuracy in practical applications of capacity estimation across

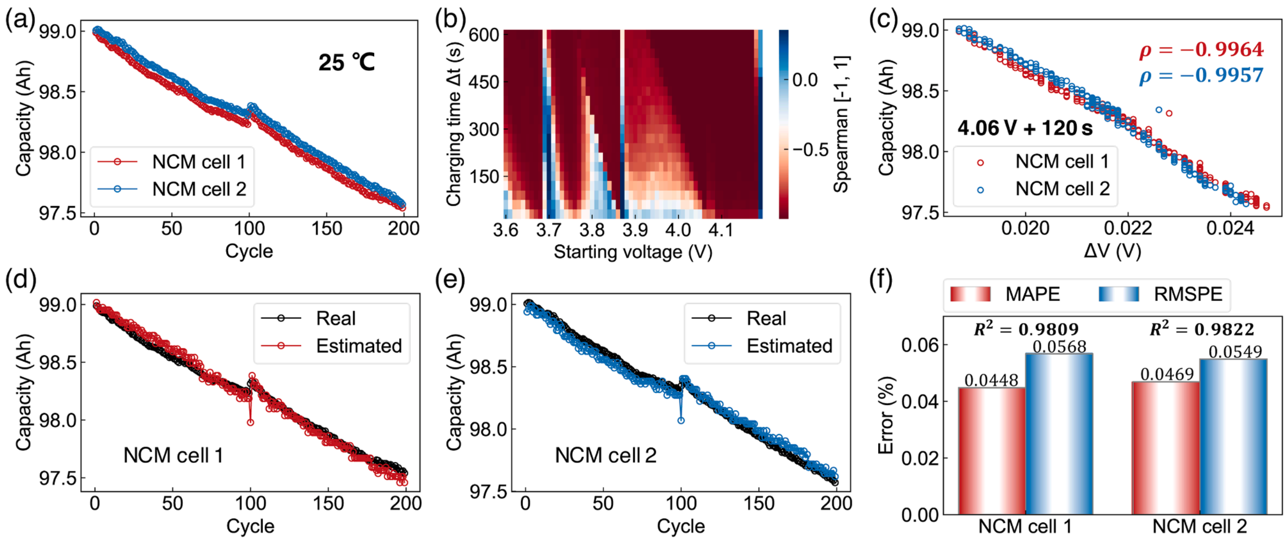

Fig. 9. Capacity estimation results for two NCM cells. (a) Capacity degradation trajectory. (b) Health feature extraction. (c) The mapping between the selected features and capacity. Capacity estimation results for (d) NCM cell 1 and (e) NCM cell 2. (f) Capacity estimation errors of two cells.

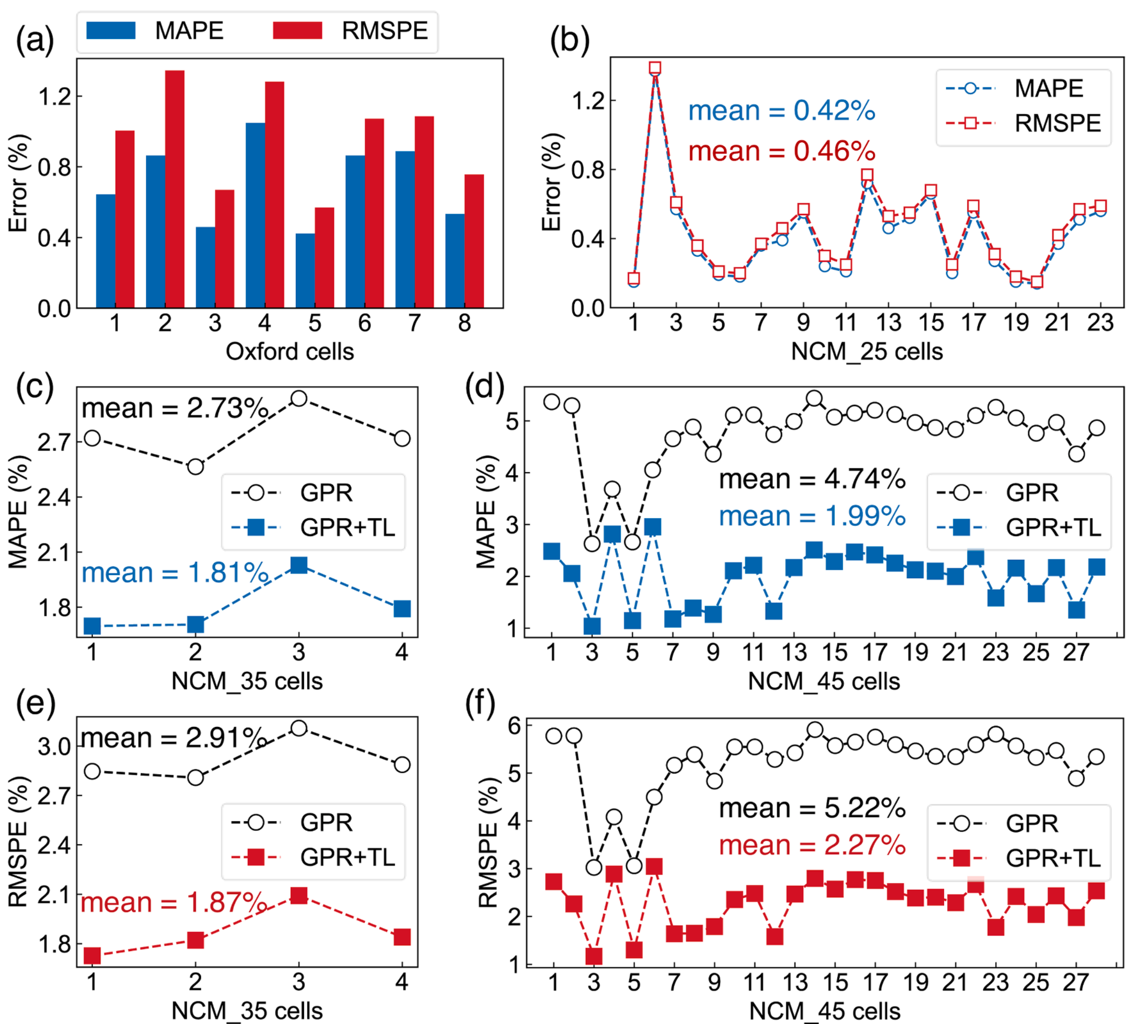

Fig. 10. Battery capacity estimation results under constant-current charging scenario. (a) Oxford dataset. (b) NCM_25 cells 1–23 (25 °C). MAPEs of temperature transfer estimation for (c) NCM_35 cells 1–4 (35 °C) and (d) NCM_45 cells 1–28 (45 °C). RMSPEs of temperature transfer estimation for (e) NCM_35 cells 1–4 and (f) NCM_45 cells 1–28.

temperature domains. In conclusion, based on a larger number of battery samples, the above results demonstrate the high generalization and transfer ability of the proposed method.

In summary, this study highlights the capability of data-driven methods to infer battery capacity under different charging modes (particularly non-constant current), operating temperatures, and battery chemistries. In practice, accurate capacity estimation forms the basis for battery remaining useful life (RUL) prediction and degradation trajectory forecasting [58,59]. Time-series-based techniques [60] or sequence-to-sequence models [59] can be used to establish mappings between historical capacity sequences and future capacity sequences for enhancing confidence in battery maintenance. Furthermore, by delving deeper into the relationship between differences in health features and battery aging, it is expected to achieve early prediction and classification of battery end-of-life (EOL) [50,61]. More broadly, due to the similarity in principles (finding mappings between partial charge curves and battery capacity) and the generality of data-driven methods, the proposed approach also exhibits tremendous prospects for further application in lithium-sulfur batteries [62] and lithium metal batteries [60].

4. Conclusions

For non-constant current charging scenarios of lithium-ion batteries, classical capacity estimation methods developed based on the constant-current charging process may not be applicable. In this context, this work proposed a fast and transferable data-driven model for battery capacity estimation during non-constant current charging. It achieved fast and accurate capacity estimation using a single health feature collected at 3 min in the high voltage region combined with GPR. The model constructed based on 25 °C can achieve transfer estimation using early cycle data from a new temperature environment. Using one cell for training, the average capacity estimation error (MAPE) for the other three cells was 0.65 %. The R^2 scores were all greater than 0.98. When extrapolating to low-temperature (10 °C) and high-temperature (40 °C) operating conditions, the accuracy of capacity estimation can be significantly improved using the proposed transfer strategy. Compared to the cases without transfer learning, the estimation errors were reduced by more than 30 %. The comparison with three classical methods showed that the proposed method improved the capacity estimation accuracy by 69 %, 82 %, and 68 %, respectively. Meanwhile, the proposed method has significant advantages in terms of sampling time, computational complexity, and practicality. Furthermore, some additional experiments highlighted the generality of the proposed approach for different battery chemistries and charging protocols.

CRediT authorship contribution statement

Chuanping Lin: Software, Writing – original draft, Formal analysis, Data curation, Visualization, Validation. **Jun Xu:** Conceptualization, Methodology, Investigation, Writing – review & editing, Supervision, Funding acquisition. **Jiayang Hou:** Software, Formal analysis, Writing – review & editing. **Delong Jiang:** Software, Investigation, Writing – review & editing. **Ying Liang:** Software, Formal analysis, Writing – review & editing. **Xianggong Zhang:** Data curation, Visualization, Validation, Writing – review & editing. **Enhu Li:** Data curation, Visualization, Validation, Writing – review & editing. **Xuesong Mei:** Resources, Supervision, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ensm.2023.102967.

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